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## Masterblend Feeding Chart

*Per-gallon feeding by crop and stage · distilled from Guide 4 Part 1*

The base ratio is **2 : 1 : 2** (Masterblend 4-18-38 : Epsom salt : Calcium nitrate 15.5-0-0). Adjust concentration, not ratio. The one canonical exception: peppers at fruiting take a magnesium bump (the 2.4 g/gal rule).

### Pepper — Full Cycle

\*Guide 4 §1.4.2\*

| Week       | Stage                     | MB g/gal   | Epsom g/gal | Cal-N g/gal | Target EC      | pH             |
|------------|---------------------------|------------|-------------|-------------|----------------|----------------|
| 1-3        | Seedling                  | 1.2        | 0.6         | 1.2         | 0.8-1.0        | 5.8-6.0        |
| 3-6        | Vegetative                | 2.4        | 1.2         | 2.4         | 1.4-1.8        | 5.8-6.0        |
| 6-10       | Pre-flower                | 2.8        | 1.4         | 2.8         | 1.8-2.2        | 5.8-6.2        |
| <b>10+</b> | <b>Fruiting (Mg bump)</b> | <b>3.0</b> | <b>2.4</b>  | <b>3.0</b>  | <b>2.2-2.8</b> | <b>6.0-6.2</b> |

### Tomato — Full Cycle

\*Guide 4 §1.4.1\*

| Week          | Stage                 | MB g/gal | Epsom g/gal | Cal-N g/gal | Target EC | pH      |
|---------------|-----------------------|----------|-------------|-------------|-----------|---------|
| 1-2           | Germination           | —        | —           | —           | 0.0       | 6.0     |
| 2-4           | Seedling              | 1.2      | 0.6         | 1.2         | 0.8-1.0   | 5.8-6.0 |
| 4-7           | Early veg             | 2.4      | 1.2         | 2.4         | 1.4-1.8   | 5.8-6.0 |
| 7-10          | Late veg / transition | 2.8      | 1.4         | 2.8         | 1.8-2.2   | 5.8-6.2 |
| 10-14         | First truss flowering | 3.0      | 1.5         | 3.0         | 2.0-2.4   | 5.8-6.2 |
| 14-20         | Heavy fruiting        | 3.6      | 1.8         | 3.6         | 2.4-3.0   | 6.0-6.2 |
| Final 1-2 wks | Flush / finish        | 0.5      | 0.3         | 0.5         | 0.5-0.8   | 6.0     |

**The canonical 2.4 g/gal Epsom rule.** Pepper fruiting Epsom is 2.4 g/gal — *not* 1.5. Peppers lift more Mg through fruiting than any other common hydroponic crop. Hold the bump from first fruit set through harvest. This supersedes the 1.5 g/gal figure in upstream Masterblend tables (which average tomatoes and peppers and undercount peppers). Tomato Mg bump is smaller — 1.5-1.8 g/gal across weeks 14-20.

**Mixing Order — A → Epsom → CaN-last**

\*Guide 4 §1.3\*

| Step | Action                                              | What happens                                                         |
|------|-----------------------------------------------------|----------------------------------------------------------------------|
| 1    | Fill reservoir with treated water at working volume | Bicarbonates already neutralized (see Cheat Sheet 3)                 |
| 2    | Dissolve Masterblend 4-18-38 fully                  | Solution turns pale green                                            |
| 3    | Dissolve Epsom salt fully                           | Solution remains clear                                               |
| 4    | Dissolve Calcium nitrate <b>last, fully</b>         | Solution stays clear — any cloudiness = precipitation, du<br>restart |

**WARNING**

**Never mix calcium nitrate directly with Masterblend concentrate** or with un-dissolved Epsom. Free phosphate + free calcium = insoluble calcium phosphate, which precipitates onto roots within minutes. Dissolve in the order above, in a *diluted* reservoir, fully mixed between steps.

**5-Gallon Batch Reference**

\*Guide 4 §1.4.4\*

| Use case                          | MB          | Epsom       | Cal-N       | Target EC      |
|-----------------------------------|-------------|-------------|-------------|----------------|
| Seedlings / cuttings              | 6 g         | 3 g         | 6 g         | 0.8-1.0        |
| Vegetative                        | 12 g        | 6 g         | 12 g        | 1.4-1.8        |
| <b>Fruiting (pepper, Mg bump)</b> | <b>15 g</b> | <b>12 g</b> | <b>15 g</b> | <b>2.2-2.8</b> |
| Fruiting (tomato)                 | 15 g        | 7.5 g       | 15 g        | 2.0-2.4        |
| Heavy fruiting (tomato)           | 18 g        | 9 g         | 18 g        | 2.4-3.0        |

**LOCAL TIP**

**A 0.1 g jeweller's scale is non-negotiable.** The difference between 1.2 g and 2.4 g of Epsom on a single gallon is the difference between healthy mid-leaf color and interveinal yellowing on the lower trusses. A \$15 scale from ACE Hardware in Sierra Vista pays for itself.